Name ___________________________

Unit 10 - Stoichiometry
Essential Knowledge and Guided Notes

Textbook: CK Chemistry Chapter 12 - Stoichiometry
https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/

Helpful Videos and Practice: Posted on CANVAS

Essential Knowledge:

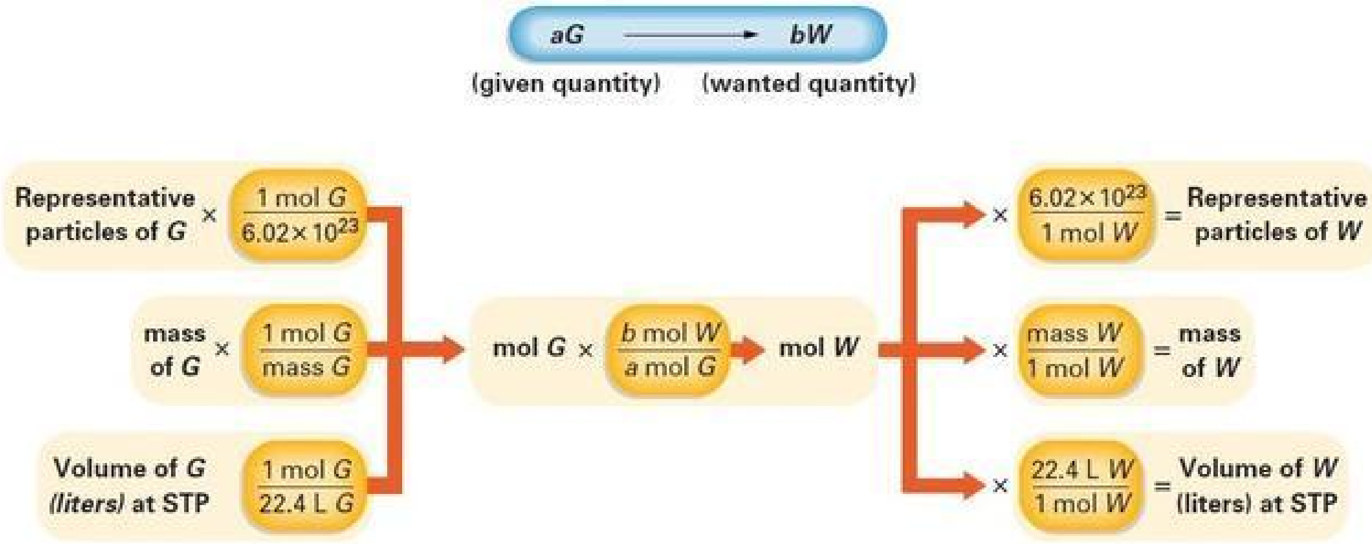
1. A Balanced Chemical Equation provides the same kind of quantitative information that a recipe does. Chemists use balanced equations as a basis to calculate how much reactant is needed and how much product is formed in a reaction. In a balanced equation total grams of the reactant(s) = total grams of the products.
2. Stoichiometry involves quantitative relationships. Stoichiometric relationships are based on mole quantities in a balanced equation.
3. Mole Ratios are used to convert between moles of reactant and moles of product, between moles of products, or between moles of the product. In a typical stoichiometric problem, the given quantity is first converted to moles. Then the mole ratio from the balanced equation is used to calculate the moles of the wanted substance. Finally, the moles are converted to any other unit of measurement related to the mole unit.
   a. 1 mole of gas at STP =22.4 L
   b. 1 mole of a substance = 6.02×10^{23} particles
   c. 1 mole of an element or compound = Molar Mass in Grams (use your Periodic Table to find this!)
4. Using these relationships, you should be able to perform stoichiometric calculations involving the following relationships:
   a. mole-mole
   b. mass-mass
   c. mole-mass
   d. mass-volume
   e. mole-volume;
   f. volume-volume
   g. mole-particle
   h. mass-particle
   i. volume-particle
   j. heat - mole
   k. heat - mass
   l. heat - volume
   m. heat - particle
5. The steps in a basic stoichiometry problem are always the same:
   a. Write and balance the chemical equation involved.
   b. Convert grams/liters/particles of the given substance to moles (This is always a division)
   c. Set up a mole ratio from the balanced equation with the coefficient of the wanted on the top and the coefficient of what you are given to find on the bottom.
   d. Convert moles of what you want to find to grams/liters/particles/heat (This is always multiplication)
6. Stoichiometry predicts the theoretical yield mathematically. Scientists get an experimental or actual yield when carrying out a reaction in a lab. You will need to be able to calculate the percent yield of a reaction given the experimental yield and after having calculated the theoretical yield using stoichiometry.

$$\%\ yield = \frac{experimental\ yield}{theoretical\ yield} \times 100$$

7. Some stoichiometry problems do not identify the limiting reagent (or reactant). The limiting reactant is the one that will be in insufficient quantity and will limit the amount of product formed. You should be able to identify the limiting reactant and use it to calculate the correct

yield of the experiment. This is done by seeing which of the given reactants gives the smallest yield of product. This will be the limiting reactant.

Reference:



Convert to Moles (divide) | Mole Ratio from Equation | Convert to Desired Unit (multiply)

Working a Stoichiometry Problem

6.50 grams of aluminum reacts with an excess of oxygen. How many grams of aluminum oxide are formed?

$$4\ Al + 3\ O_2 \rightarrow 2Al_2O_3$$

$$\frac{6.50\ \cancel{g\ Al}}{} \times \frac{1\ \cancel{mol\ Al}}{26.98\ \cancel{g\ Al}} \times \frac{2\ \cancel{mol\ Al_2O_3}}{4\ \cancel{mol\ Al}} \times \frac{101.96\ g\ Al_2O_3}{1\ \cancel{mol\ Al_2O_3}} = ?\ g\ Al_2O_3$$

$$6.50 \times 2 \times 101.96 \div 26.98 \div 4 = 12.3\ g\ Al_2O_3$$

U10 Lesson #1: Mole/Mole Stoichiometry

If you are a chemist who needs to synthesize materials, you want to be sure you can predict how much product you can provide to your client. Fortunately, this is as simple as figuring out how much cheese and how many crackers you need for a cheese and cracker sandwich.

In a chemical equation, the ratio of the coefficients (called the mole ratio) is used to make the prediction.

Cheese and Cracker Stoichiometry Problems - Proof of Method

Balanced equation:

1 cheese + 2 crackers → 1 sandwich

Ratios from this Equation:

1 cheese / 2 crackers	2 crackers / 1 sandwich	1 sandwich / 1 cheese

and the inverse of each.

Example #1: How many crackers are needed to make 5 sandwiches? Pick the correct ratio.

Given **Ratio** **Want**

5 sandwiches x ____________________ = Crackers

Example #2: How many sandwiches can be made from 14 crackers?

Given **Ratio** **Want**

x ________________ =

The limiting reactant or limiting reagent is a reactant in a chemical reaction that determines the amount of product that is formed. Practically - It runs out first!!!

Example #3

With 4 slices of cheese and 10 crackers, which is the limiting reactant? Show a calculation

Given **Ratio** **Want**

4 slices of cheese x _______________ = _________ sandwiches

Given **Ratio** **Want**

10 crackers x ________________ = _________ sandwiches

How many sandwiches can be made? ______Which is the limiting reactant? ____________

Why?_________________________

What do we call the reactant that is NOT the limiting reactant? ___________________________

Chemistry Stoichiometry Problems

$$2\ H_2 + O_2 \rightarrow 2\ H_2O$$

Mole ratios from this equation:

$\frac{\textbf{2 mol } H_2}{\textbf{1 mol } O_2}$ $\frac{\textbf{2 mol } H_2}{\textbf{2 mol } H_2O}$ $\frac{\textbf{1 mole } O_2}{\textbf{2 mol } H_2O}$ **and inverse**

Example # 1: How many moles of oxygen gas (O_2) are needed to make 15 moles of H_2O?

Given		Ratio		Want
15 mol H_2O	x	______________	=	moles of O_2

$$2\ Al + 3\ CuCl_2 \rightarrow 2\ AlCl_3 + 3\ Cu$$

Mole ratios from this equation:

_________ _________ _________ __________ __________ ___________ **and inverse**

Example #2: How many moles of Al will react with 6.0 moles of $CuCl_2$?

Given **Ratio** **Want**

Example #3: How many moles of Cu will form from the reaction of 3.0 moles of Al and excess $CuCl_2$?

Given **Ratio** **Want**

Example #4: How many moles of $AlCl_3$ will form from 0.25 moles of Al?

Given **Ratio** **Want**

Example #5: How many moles of Al are needed to form 7.0 moles of Cu?

Given **Ratio** **Want**

U10 Lesson #2: Unit/Unit Stoichiometry

1. **Balance the Equation**
2. **Convert the given unit to Moles (/)**
3. **Set up the correct Mole Ratio to convert moles of the given to moles of the wanted**
4. **Convert Moles of the wanted to the desired unit. (x)**

Example #1:

How many grams of Al are needed to completely react with 135 grams of Fe_2O_3?

What is Given? ____________________________ What is Wanted? ___________________

Balanced Equation:

$$Fe_2O_3 + 2\ Al \rightarrow 2\ Fe + Al_2O_3$$

Molar Mass of Fe_2O_3 = 2(55.85) + 3(16.00) = 159.70 grams
Molar Mass of Al = 26.98 grams
Molar Mass of Fe = 55.85 grams

Given (g)		**Convert to Moles**		**Ratio**		**Convert to Grams**		**Want**
135 g Fe_2O_3	x	_______________	x	________	x	_______________	=	__________g Al

Example #2:

How many grams of Cu can be produced by 4.55 grams of Al when it reacts with excess $CuCl_2$ to form Cu and $AlCl_3$?

Balanced Equation:

Molar Mass of Cu = 63.55 grams
Molar Mass of Al = 26.98 grams

What is Given? ___________________________ What is wanted? ________________________

Given (g) Convert to Moles Ratio Convert to Grams Want

Example #3:

How many liters at STP of NH_3 will be produced from 1.204×10^{25} molecules of $H_2(g)$?

What is Given? ___________________________ What is wanted? ________________________

Balanced Equation:

Given (g) Convert to Moles Ratio Convert to Liters Want